Capacity feasible clustering with neutral atoms

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We built an atom-efficient clustering formulation and a verified BLaDE embedding workflow. Reduced assignment uses 4, 20 and 45 atoms instead of 14, 45 and 80 with capacity slack, saving 71%, 56% and 44%. BLaDE lowers pair-interaction error relative to our custom placement in the tested medium and large cases. The remaining failure point is faithful encoding: soft balance can alter the intended answer, and atomic geometry can distort the interactions further. Our contribution is a compact implementation plus controlled evidence locating these losses. All quantum results are noiseless emulations.

Assignment encodings

Each customer must enter one vehicle group. We minimize pairwise Euclidean distance within groups, subject to a common vehicle capacity. Unused vehicles are allowed; routing is evaluated separately after clustering. Binary indicators give the constrained model:

$$min_x F(x) = \sum_{k=1}^{K} \sum_{i<j} c_{ij} x_{ik} x_{jk}, \sum_k x_{ik} = 1, L_k = \sum_i d_i x_{ik} \le Q.$$

The one-hot baseline uses one bit per customer and vehicle. Reduced assignment omits the last vehicle indicator. Valid codes below select vehicles in order; the all-zero reduced code selects the last vehicle.

Vehicles	One hot valid codes	Reduced valid codes
2	10, 01	1, 0
3	100, 010, 001	10, 01, 00
4	1000, 0100, 0010, 0001	100, 010, 001, 000

$$P_{onehot} = A \sum_i \left(\sum_k x_{ik} - 1 \right)^2, x_{iK} = 1 - \sum_{k<K} y_{ik}.$$

Reduced assignment excludes multiple active bits using a bit-removal energy bound for its penalty. Balance weight matches mean objective and demand-pair coupling scales. Full one-hot balance uses a feasible-energy bound for assignment. Linear substitution remains quadratic:

$$K = 2 : P_i = 0 ; K = 3 : P_i = M a_i b_i ; K = 4 : P_i = M \left(a_i b_i + a_i c_i + b_i c_i \right).$$

Capacity slack versus load balance

$$E_{slack} = F + P_{assign} + B \sum_k \left(L_k + s_k - Q \right)^2.$$

$$E_{balance} = F + P_{assign} + \mu \sum_k \left(L_k - Q \right)^2.$$

Slack is a nonnegative integer encoded in bounded binary bits. Every feasible load can choose spare capacity to make its penalty zero; an overload cannot. In the main benchmark, assignment and capacity penalties exceed the cost of a demand-feasible certificate, so every intended ground state is feasible and original-optimal. Balance needs no slack but penalizes valid unequal loads too: at capacity 7, loads (5,5) and (6,4) have zero slack penalty but balance penalties of 8 and 10 times the balance weight. Loads (8,2) cost 26 times that weight but remain possible if the objective compensates.

Encoding	Small atoms	Medium atoms	Large atoms
Assignment plus capacity slack	14	45	80
Full one hot plus balance	8	30	60
Reduced assignment plus balance	4	20	45

Balance is a heuristic. On valid assignments its capacity-dependent terms are constant at fixed weight. A separate 100-problem audit found four cases where no balance weight recovers the constrained optimum. [1]

Instance generation and verification

We generate seeded customer positions uniformly in a 100 by 100 square, with depot (50,50), then integer demands. Starting at seed 2000, exact screening rejects infeasible draws and keeps only cases where capacity changes the optimal clustering. The benchmark is deliberately capacity-active, not an unfiltered random population. [2]

Tier	Problems	Customers	Vehicles	Capacity	Demand range
Small	10	4	2	7	1 to 4
Medium	5	10	3	20	1 to 8
Large	5	15	4	30	1 to 10

All 20 instances and demand-only certificates validate. Exact enumeration and dynamic programming prove both references; each case has one optimal clustering modulo vehicle relabeling. All ten small cases agree with 20 CPLEX solves (30-second limit, one thread, seed 0, zero MIP gaps). Exact screening scales exponentially.

$$\rho = \frac{\sum_i d_i}{K\,Q}, \delta = \frac{d_{max}}{Q}.$$

Here delta denotes largest-demand ratio. Load-ratio ranges: small 0.786–0.929, medium 0.700–0.900, large 0.733–0.867. Largest-demand ratios: small 0.429–0.571, medium 0.400, large 0.333. Seeds, certificates and ratios accompany the saved instances.

Embedding fidelity

Each bit becomes an atom, with positive interactions including unwanted tails. Geometry requires 5 micrometers minimum separation and 38 micrometers maximum radius. A global laser cannot match arbitrary individual fields. BLaDE lowers mean pair error versus our custom placement: 1.025 to 0.534 for medium reduced and 1.319 to 0.370 for large reduced. Search budgets differ. [2, 3]

$$U_{ij} = C_6 / \| r_i - r_j \|^6, \varepsilon_J = \frac{\| U - \alpha J \|_F}{\| \alpha J \|_F}, \alpha = \frac{\sum_{i<j} U_{ij} J_{ij}}{\sum_{i<j} J_{ij}^2}.$$

BLaDE mean pair error	Small	Medium	Large
Reduced assignment plus balance	0.204	0.534	0.370
Assignment plus capacity slack	0.497	0.993	1.201

Only off-diagonal pairs enter the norm. Lower is better; this is coefficient error, not quantum-state fidelity. Three layouts are averaged within each of 10 small, 5 medium and 5 large problems. Preparation failures remain in these geometry averages.

Pair distortion changes the preferred answer

We solve the intended QUBO, replace only its pairs with geometric interactions while preserving individual fields and scale, then test the actual endpoint fields. The table counts layouts with a feasible minimum, not quantum probabilities. [3]

Encoding	Problems	Layouts	Intended	Pairs changed	Endpoint
Small slack	10	60	60 of 60	0 of 60	1 of 60
Medium reduced	5	10	10 of 10	0 of 10	0 of 10
Medium slack	5	10	10 of 10	0 of 10	0 of 5 available

Small and 20-bit checks enumerate all states and ties; 45-bit checks use optimization bounds. One reduced intended model is already suboptimal. A later BLaDE audit finds no feasible minima in 80 pair-only models and 72 endpoints across five medium problems; eight endpoints are unavailable. Repeated settings are not independent instances.

Matched emulations across encoding methods

To isolate pair distortion in dynamics, we ran a new comparison on the same three small problems, seeds 2002, 2006 and 2012. Each encoding receives one BLaDE layout with fixed placement seed. Within each exact/geometric pair, the initial state, four-microsecond analytic pulse, individual fields and energy scale are identical. Only pair interactions change. Penalties use instance data and feasible certificates, not known optimal answers. [4]

Encoding	Atoms	Feasible exact	Feasible geometry	Optimal exact	Optimal geometry
Capacity slack	14	7.81%	1.10%	3.90%	0.55%
Full one hot plus balance	8	21.61%	10.55%	16.00%	9.67%
Reduced plus balance	4	36.05%	24.00%	29.03%	16.38%

Feasibility and original-clustering optimality decrease in all nine pairs. All 18 distributions pass timestep-halving checks below 0.0001 total variation; two eight-bit arms also agree with independent adaptive integration. All nine polynomial expansions were checked on every bitstring and all 18 distributions were independently rescored. The run completed in 75.3 seconds with no cloud jobs. These are computed state probabilities, not shot fractions.

The result identifies pair distortion as a bottleneck in these cases, but not the only limitation. Both balance encodings already have an infeasible intended minimum on one problem. Its reduced geometric model restores an optimal minimum yet still samples worse under the fixed pulse. Ground-state correctness and dynamical preparation are distinct. Full one-hot medium and large emulations remain untested. Individual-field controls are emulator diagnostics, and the new analytic pulse and conservative full one-hot penalties differ from historical settings.

The historical medium reduced cloud control supports the same contrast: 37.025% feasible with exact interactions versus 4.020% with geometric pairs, with 20,000 shots per condition. Finer local simulations give 36.7942% versus 3.8912%. This remains one medium problem. Forty-five-atom slack runs are poor even with intended pairs, and their numerical refinement remains incomplete. [5]

Classical sampling remains the stronger baseline

The main benchmark uses a global four-microsecond interpolated pulse: near-zero drive rises to its device maximum, holds, then returns near zero; detuning sweeps from negative twice that amplitude through zero to the fitted endpoint. Three layouts with 2,000 shots each give 6,000 samples per problem. Classical controls use the same 10/5/5 problems and sample counts, not equal compute. [2]

Metric	BLaDE reduced	Balanced random	Uniform
Small feasible	38.63%	70.11%	31.21%
Small optimal	28.675%	29.6967%	12.72%
Medium feasible	3.6833%	48.4767%	19.1333%
Medium optimal	0%	0.0500%	0.0067%
Large feasible	0.1633%	55.3167%	12.5567%
Large optimal	0%	0%	0%

Balanced random equalizes customer counts, not demand, and does not guarantee capacity. Uniform assigns each customer independently. All outcomes remain in the denominator; there is no repair or postselection. Zero observed optima do not establish zero probability. State vectors cover up to 20 atoms. Larger runs use MPS bond dimension 64, with selected checks at 128; some observables disagree, so convergence is unresolved. Noise and modulation are off.

The main two-method study completed 209 of 240 primary conditions; 31 failed preparation. Transfer and accuracy tests bring the historical total to 301 completed runs, 47 preparation failures and 602,000 shots. The new local probability calculations are separate and add no shots.

Pulse tuning

We tuned duration, drive and detuning on fourteen-atom slack problems: 944 trials, 34 winners and 680 case/interaction evaluations. Winners trained on three problems and were evaluated unchanged on two validation and five evaluation problems. Previously inspected cases make these retrospective tests. [5]

Complete-answer tuning requires an optimal assignment and correct slack bits. The table uses known-answer-guided penalties; this is a diagnostic separate from the answer-independent comparison on page 3.

Training interactions	Evaluation interactions	Complete optimum probability
Intended	Intended	11.8–23.7%
Intended with identical pulses	Geometric	Below reporting floor
Geometric	Geometric	Up to 0.168%

Ranges are three search-seed means over five evaluation problems. The 0.1% reporting floor is not an upper bound. Original penalties yield 1.33–2.26% under intended interactions. Known-answer penalties, ideal individual fields and some overlong pulses limit practical applicability.

Assignment-focused tuning reached 41.5–49.4% optimal assignments but only 0.221–7.63% complete answers. Changing the tuning objective also changed the pulse menu, preventing isolated attribution. One random-search control was competitive; direct geometric tuning remained weak.

Track A metrics and best feasible gaps

The results CSV covers all 240 primary conditions, including 31 preparation failures, with per-instance/encoding/embedder/seed feasibility, reference-hit probability, atoms, pair error, reference status, optimum count, instance ratios and both gaps. Missing or no-feasible-shot gaps are undefined.

$$gap = 100(C - C^*)/C^*, \varepsilon = 0, \tau_{abs} = 10^{-8}.$$

Reduced BLaDE	Small	Medium	Large
Best observed model gap	0%	37.66%	81.90%
Best observed CVRP gap	0%	4.30%	16.19%
CVRP reference hit rate	28.6567%	0%	0%

Gaps are tier minima, not averages or necessarily the same shot. CSV also gives the CVRP gap of the best-model shot. Clusters are routed exactly by Held–Karp. Hit rates count feasible raw shots within tolerance, modulo vehicle labels; no repair. Model hit rates: page 3.

What worked and what did not

Our workflow saves atoms and improves interaction fit. Encoding fidelity remains the failure point: balance can change the intended optimum, geometry distorts pairs, and finite pulses can miss a correct ground state. This supports a useful strategy, not quantum advantage or proven optimal placement.

Next, certify models and inspect low-energy physical candidates before tuning. Full one-hot medium tests and larger numerical convergence remain open.

Evidence and reproducibility

[1] Encoding audit and 100-instance screening. [2] Scaling study, certificates, metrics and controls. [3] Classical embedding audit and BLaDE review. [4] Matched encoding controls, 18 distributions and independent scores. [5] Quantum comparison and pulse tuning. Sources and hashes accompany the report. Earlier helper-atom and recursive tests remain separate in the methods supplement.

AI disclosure. Codex assisted with derivations, code, simulations, analysis and writing. The team remains responsible for review and submission; human review is not implied.